

# PHISHING EMAIL DETECTION & AWARENESS SYSTEM

**Future Interns  
Cyber Security Internship**

**Task 02**

Prepared By

Mohammad Aliraza Muphid Koke  
Bachelor of Engineering (Computer Engineering)



# EXECUTIVE SUMMARY

This report presents the analysis of five email samples to identify phishing attacks and improve user security awareness. Each email was examined based on sender authenticity, domain legitimacy, email content, embedded links, attachments, urgency, and social engineering techniques.

The analyzed emails were classified into three categories: Safe, Suspicious, and Phishing. The assessment demonstrates how phishing attacks commonly exploit trust, urgency, and impersonation to deceive users.

The report also provides practical recommendations and security awareness guidelines to help individuals and organizations identify phishing attempts and reduce the risk of successful email-based attacks.

# OBJECTIVE

The objective of this project is to analyze phishing email samples, identify phishing indicators, classify emails as Safe, Suspicious, or Phishing, and improve user awareness against email-based cyber threats.

The assessment focuses on recognizing malicious emails through sender verification, URL inspection, attachment analysis, and social engineering detection.

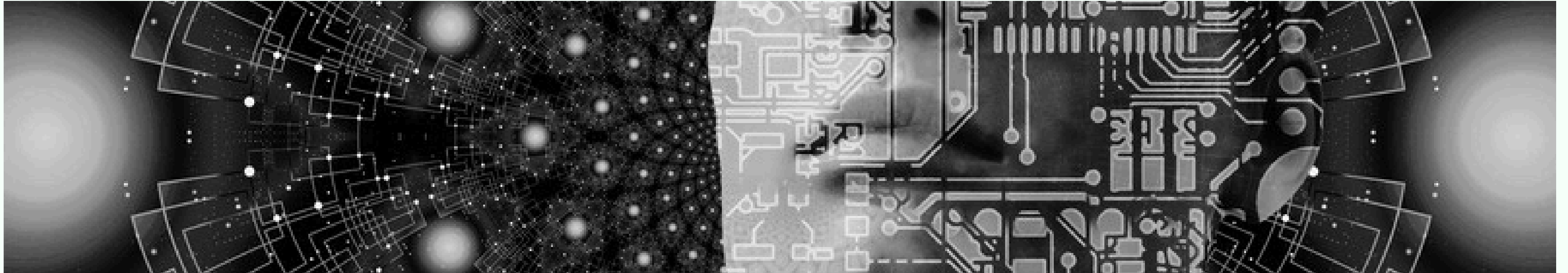
# SCOPE

## Included

- Sender verification
- Subject line analysis
- URL inspection
- Attachment analysis
- Social engineering analysis
- Email risk classification
- Security awareness recommendations

## Excluded

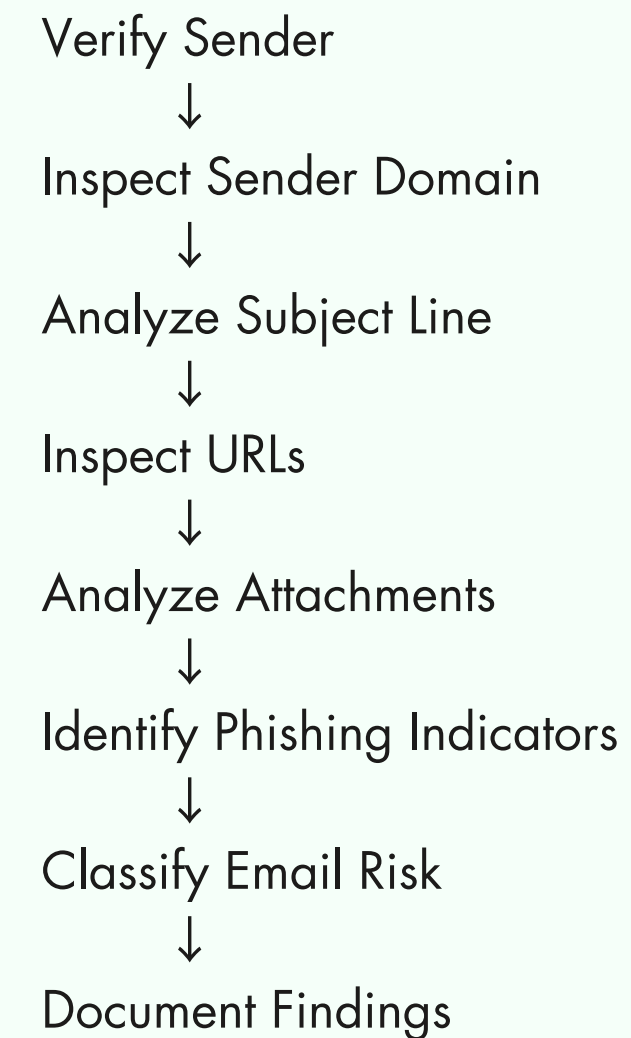
- Clicking malicious links
- Executing suspicious attachments
- Exploiting systems
- Offensive security activities



## TOOLS USED

- Kali Linux
- Visual Studio Code
- Git
- GitHub
- Browser Developer Tools
- Google Message Header Analyzer
- MXToolbox Email Header Analyzer

## Methodology



```
home > aalll > FutureInterns > FUTURE_CS_02 > evidence > emails > email_01.txt
```

```
1 From: notifications@github.com
2 To: user@example.com
3 Subject: New sign-in to your GitHub account
4
5 Hello,
6
7 We noticed a new sign-in to your GitHub account.
8
9 Time: 10 July 2026
10 Device: Firefox on Linux
11 Location: Mumbai, India
12
13 If this was you, no further action is required.
14
15 If you do not recognize this activity, please review your account security
16
17 https://github.com/settings/security-log
18
19 Thank you,
20 GitHub Security Team
21 |
```

# EMAIL SAMPLE 1

## SAFE EMAIL ANALYSIS

### Sender

notifications@github.com

### Subject

New sign-in to your GitHub account

### Key Findings

- Trusted GitHub domain
- Official security notification
- No credential request
- No suspicious attachment
- Official GitHub Security Log link

### Verdict

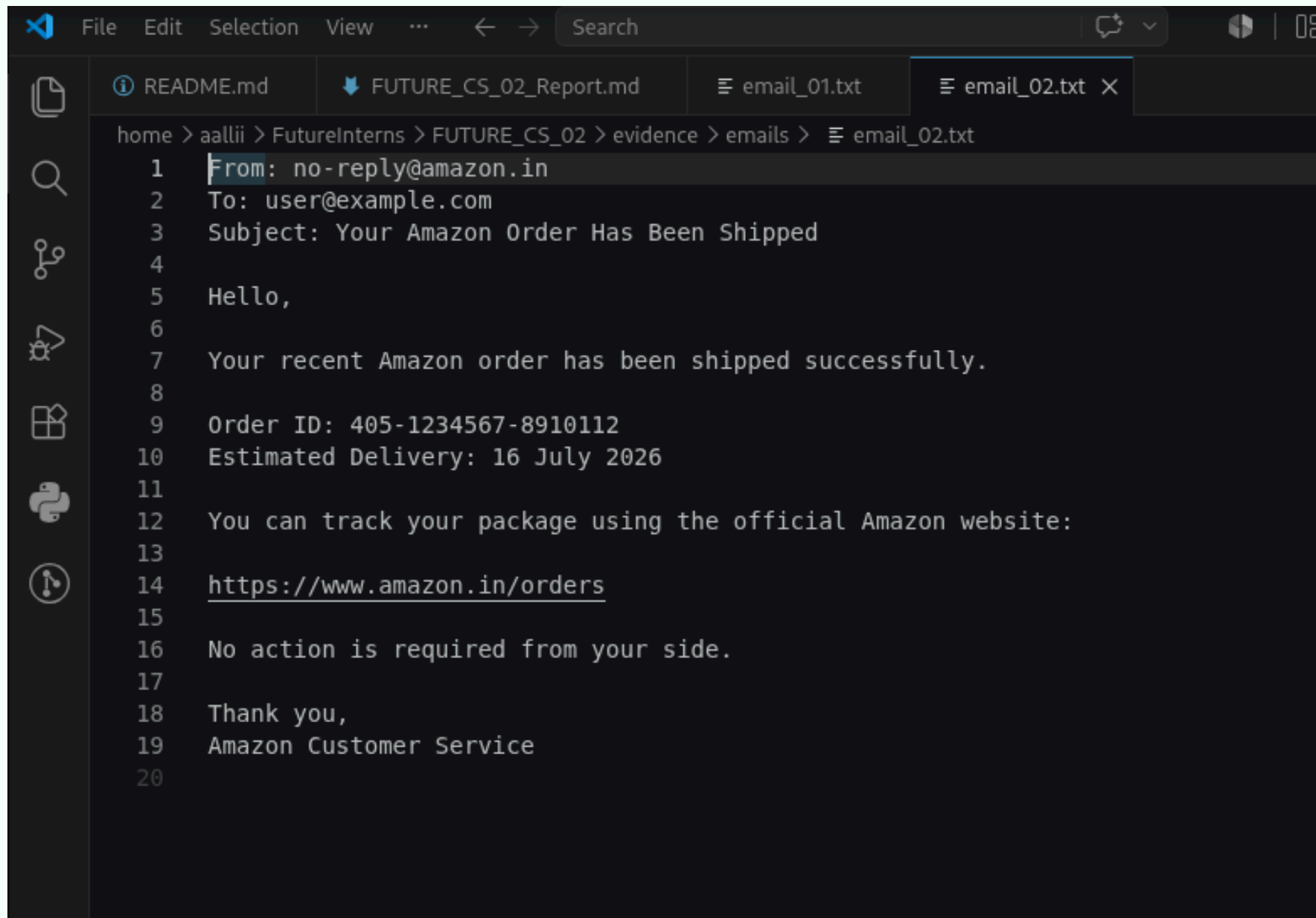
Legitimate notification from GitHub.

No phishing indicators identified.

### Classification

● Safe





```
1 From: no-reply@amazon.in
2 To: user@example.com
3 Subject: Your Amazon Order Has Been Shipped
4
5 Hello,
6
7 Your recent Amazon order has been shipped successfully.
8
9 Order ID: 405-1234567-8910112
10 Estimated Delivery: 16 July 2026
11
12 You can track your package using the official Amazon website:
13
14 https://www.amazon.in/orders
15
16 No action is required from your side.
17
18 Thank you,
19 Amazon Customer Service
20
```

# EMAIL SAMPLE 2

## SAFE EMAIL ANALYSIS

### Sender

no-reply@amazon.in

### Subject

Your Amazon Order Has Been Shipped

### Key Findings

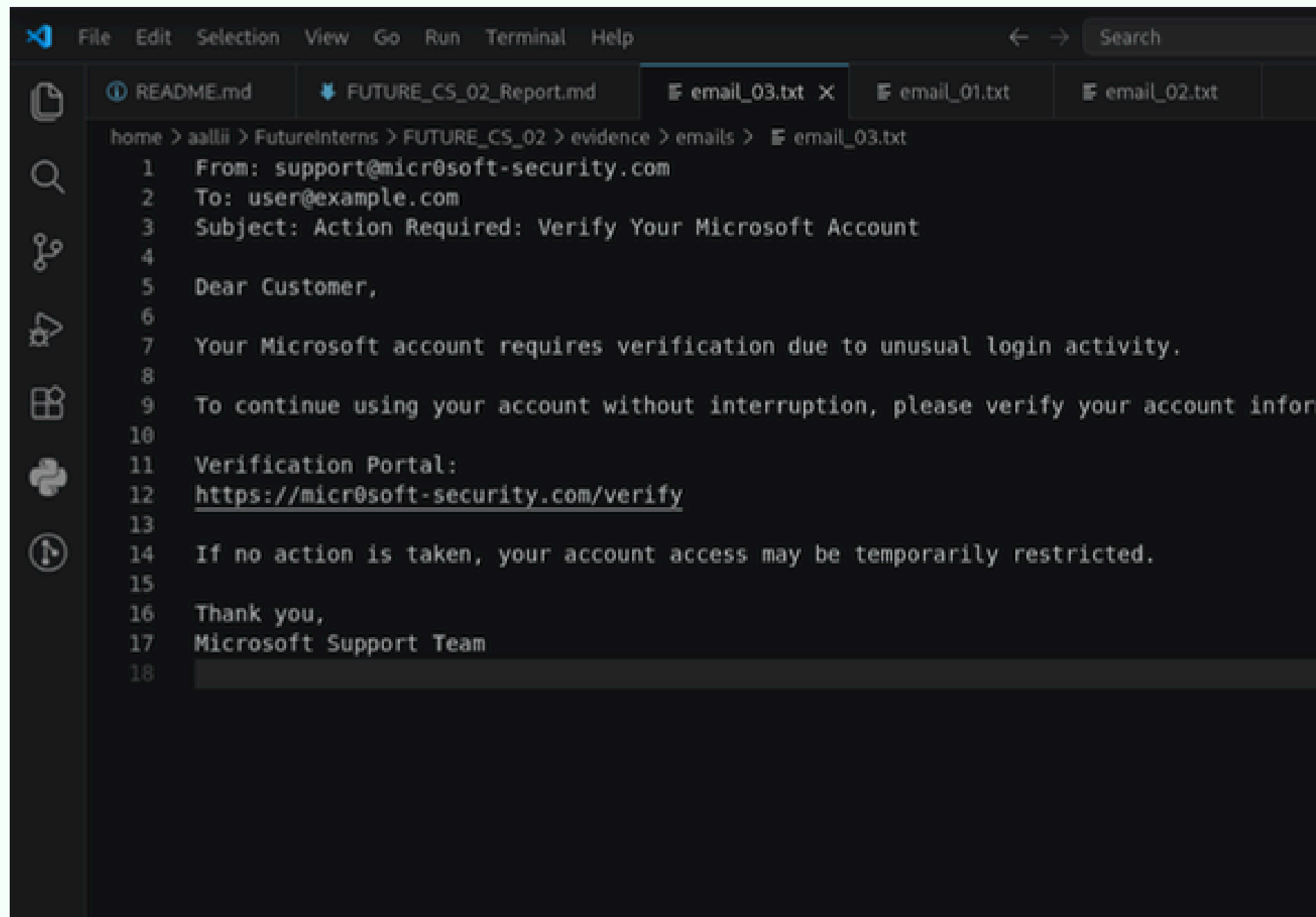
- Trusted Amazon domain
- Official order notification
- No suspicious attachment
- No credential request
- Legitimate tracking information

### Verdict

The email is a legitimate order notification from Amazon and does not contain phishing indicators or requests for sensitive information.

### Classification

 Safe



```
File Edit Selection View Go Run Terminal Help
home > aallii > FutureInterns > FUTURE_CS_02 > evidence > emails > email_03.txt
1 From: support@micr0soft-security.com
2 To: user@example.com
3 Subject: Action Required: Verify Your Microsoft Account
4
5 Dear Customer,
6
7 Your Microsoft account requires verification due to unusual login activity.
8
9 To continue using your account without interruption, please verify your account information.
10
11 Verification Portal:
12 https://micr0soft-security.com/verify
13
14 If no action is taken, your account access may be temporarily restricted.
15
16 Thank you,
17 Microsoft Support Team
18
```

# EMAIL SAMPLE 3

## SUSPICIOUS EMAIL ANALYSIS

### Sender

support@micr0soft-security.com

### Subject

Action Required: Verify Your Microsoft Account

### Key Findings

- Look-alike Microsoft domain
- Generic greeting
- Creates urgency
- Verification link is not official
- Requires manual verification

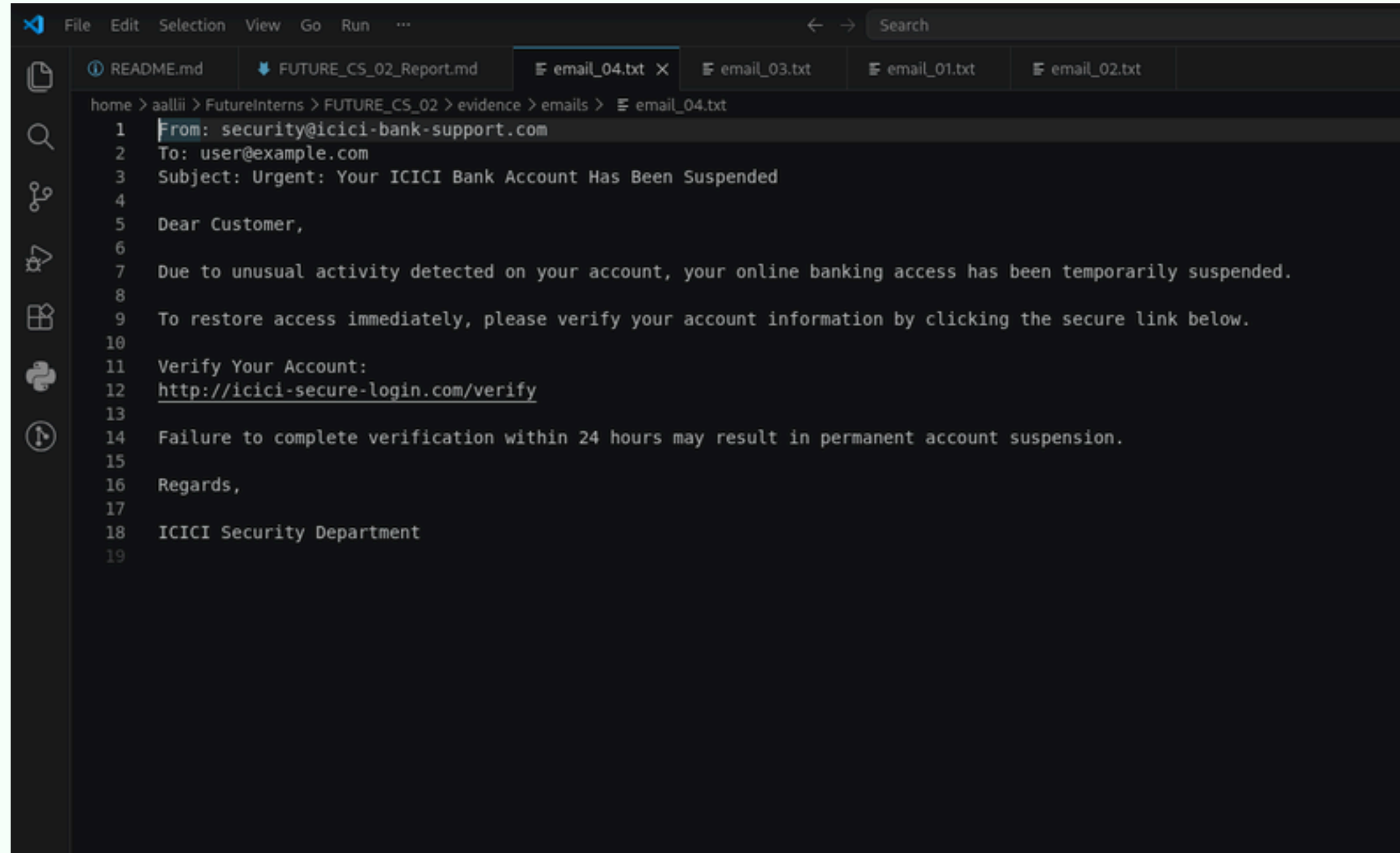
### Verdict

The sender uses a deceptive look-alike domain and encourages immediate action. Users should verify the request through Microsoft's official website before interacting with the email.

### Classification

🟡 Suspicious



A screenshot of a code editor interface with a dark theme. The top menu bar includes 'File', 'Edit', 'Selection', 'View', 'Go', 'Run', and a search bar. The file explorer on the left shows a directory structure: 'home > aallii > FutureInterns > FUTURE\_CS\_02 > evidence > emails > email\_04.txt'. The editor displays the content of 'email\_04.txt' with line numbers 1 through 19. The email text is as follows:

```
1 From: security@icici-bank-support.com
2 To: user@example.com
3 Subject: Urgent: Your ICICI Bank Account Has Been Suspended
4
5 Dear Customer,
6
7 Due to unusual activity detected on your account, your online banking access has been temporarily suspended.
8
9 To restore access immediately, please verify your account information by clicking the secure link below.
10
11 Verify Your Account:
12 http://icici-secure-login.com/verify
13
14 Failure to complete verification within 24 hours may result in permanent account suspension.
15
16 Regards,
17
18 ICICI Security Department
19
```

# EMAIL SAMPLE 4

## PHISHING EMAIL ANALYSIS

### Sender

security@icici-bank-support.com

### Subject

Urgent: Your ICICI Bank Account Has Been Suspended

### Key Findings

- Fake ICICI sender domain
- Fear-based language
- Suspicious HTTP link
- Requests banking credentials
- Attempts to steal sensitive information

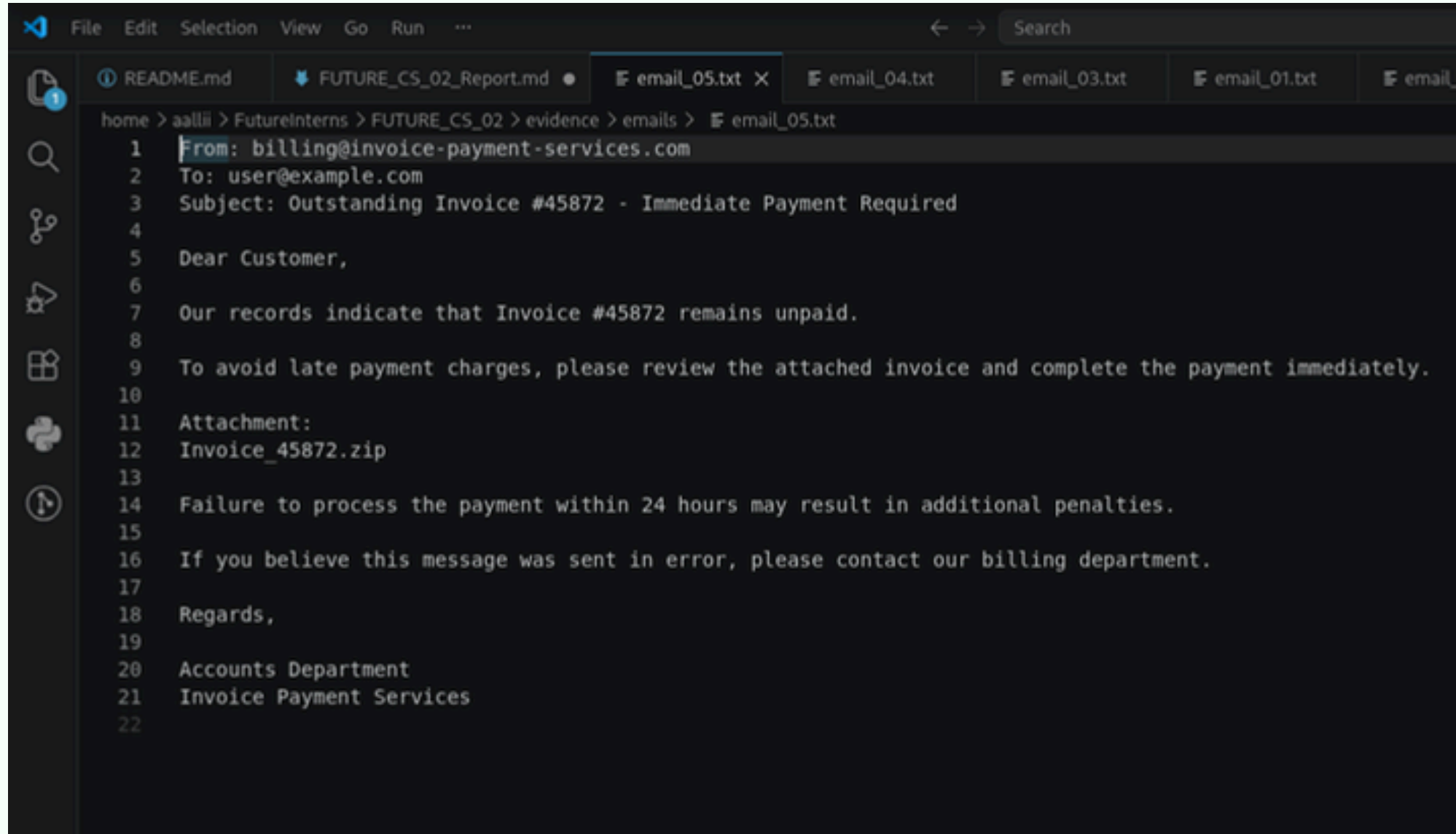
### Verdict

This email impersonates a trusted financial institution and attempts to trick users into revealing banking credentials through a fraudulent website.

### Classification

● Phishing



A screenshot of a code editor window with a dark theme. The editor has a menu bar at the top with 'File', 'Edit', 'Selection', 'View', 'Go', and 'Run'. Below the menu bar is a search bar with a magnifying glass icon and the word 'Search'. The editor shows a file explorer on the left with a tree view containing 'home > aallii > FutureInterns > FUTURE\_CS\_02 > evidence > emails > email\_05.txt'. The main editor area displays the content of 'email\_05.txt' with line numbers 1 through 22. The email content is as follows:

```
1 From: billing@invoice-payment-services.com
2 To: user@example.com
3 Subject: Outstanding Invoice #45872 - Immediate Payment Required
4
5 Dear Customer,
6
7 Our records indicate that Invoice #45872 remains unpaid.
8
9 To avoid late payment charges, please review the attached invoice and complete the payment immediately.
10
11 Attachment:
12 Invoice_45872.zip
13
14 Failure to process the payment within 24 hours may result in additional penalties.
15
16 If you believe this message was sent in error, please contact our billing department.
17
18 Regards,
19
20 Accounts Department
21 Invoice Payment Services
22
```

# EMAIL SAMPLE 5

## PHISHING EMAIL ANALYSIS

### Sender

billing@invoice-payment-services.com

### Subject

Outstanding Invoice #45872 – Immediate Payment Required

### Key Findings

- Unknown sender domain
- Suspicious ZIP attachment
- Financial pressure tactics
- Unexpected invoice
- Possible malware distribution

### Verdict

The email uses urgency and a suspicious ZIP attachment to manipulate the recipient. Opening unexpected compressed attachments may result in malware infection.

### Classification

● Phishing

# PHISHING INDICATORS

## Indicators Identified

### ● Trusted Sender Domain

- Email 1
- Email 2

### ● Urgent / Fear-Based Language

- Email 3
- Email 4
- Email 5

### ● Suspicious Attachment

- Email 5

### ● Generic Greeting

- Email 3
- Email 4
- Email 5

### ● Suspicious URL

- Email 3
- Email 4



Safe

Legitimate email with no phishing indicators.



Suspicious

Requires manual verification before interaction.



Phishing

Malicious email designed to steal credentials or distribute malware.

# RISK ASSESSMENT

## Classification

-  Safe ..... 2
-  Suspicious ..... 1
-  Phishing ..... 2

## SOCIAL ENGINEERING TECHNIQUES

- Urgency & Fear
- Fake Sender Domains
- Generic Greetings
- Credential Harvesting
- Suspicious Links
- Malicious Attachments

# DO'S & DON'TS

## • DO'S

- ✓ Verify URLs
- ✓ Enable MFA
- ✓ Report emails
- ✓ Check sender

## DON'TS

- ✗ Click unknown links
- ✗ Open ZIP files
- ✗ Share passwords
- ✗ Trust urgency

## PREVENTION GUIDELINES

- ✓ Verify sender domains
- ✓ Hover before clicking
- ✓ Never share OTPs
- ✓ Enable MFA
- ✓ Report suspicious emails

## RECOMMENDATIONS

- Awareness Training
- Email Filtering
- MFA
- Endpoint Security
- Verify Domains

## KEY LEARNINGS

- Email Analysis
- Phishing Indicators
- Social Engineering
- Documentation
- User Awareness



## **BUSINESS IMPACT**

- Reduced Credential Theft
- Reduced Malware Risk
- Improved User Awareness
- Better Security Posture

## **CONCLUSION**

This project demonstrated the practical process of identifying and classifying phishing emails using sender authenticity, email content, URLs, attachments, and social engineering techniques.

User awareness remains one of the strongest defenses against phishing attacks.